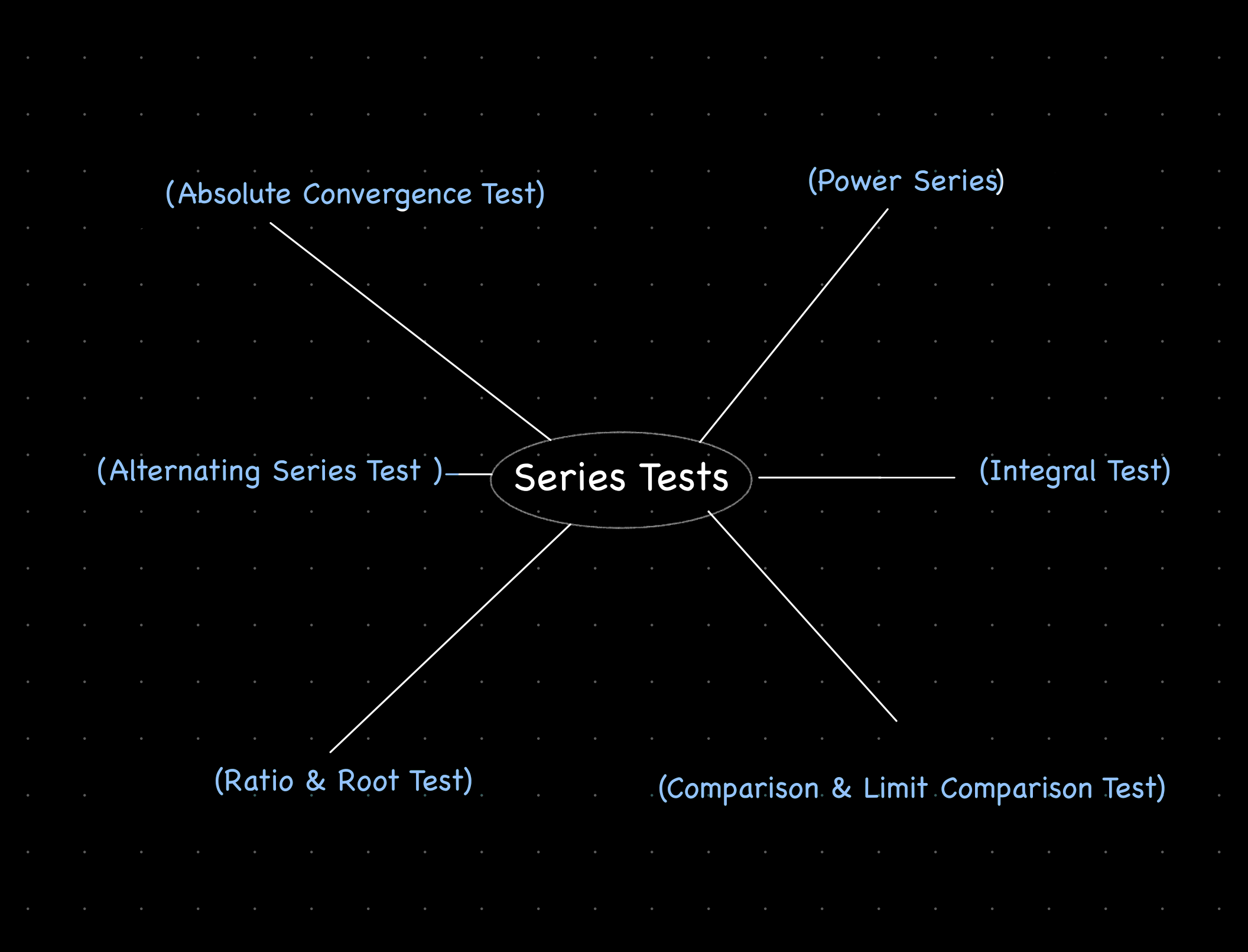

(Absolute Convergence Test)
(Power Series)
(Alternating Series Test)
Series Tests
(Integral Test)
(Ratio & Root Test)
(Comparison & Limit Comparison Test)

11.2 (The series)

- A series is the sum of the terms of a sequence: $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$
- The partial sum S_n is: $S_N = \sum_{n=1}^{N} a_n,$

and the series converges if the limit of the partial sums exists.
If the partial sums do not approach a finite (a number) limit, the series diverges.

Divergence Test (condition is: $\lim_{n\to\infty} a_n \neq 0$)

If we have $\sum_{n=1}^{\infty} a_n$

↓

Find $\lim_{n\to\infty} a_n$

If $\lim_{n\to\infty} a_n = 0$

Then the given series might be conv. or Div
to determine that
You need another test

If $\lim_{n\to\infty} a_n \neq 0$ { i.e
= number or
= ∞ or −∞
or DNE

↓

the given series is
Divergent

11.2 (The series) Continued...

Geometric Series

- A geometric series has the form: $\sum_{n=0}^{\infty} ar^n$
- where a is the first term and r is the common ratio.

- If $|r| < 1$, the series converges to:

$$S = \frac{a}{1 - r}$$ (you can only get the sum if the series converges)

- If $|r| \geq 1$, the series diverges.

P-Series

- A p-series has the form: $\sum_{n=1}^{\infty} \frac{1}{n^p}$.
- Converges if $p > 1$.
- Diverges if $p \leq 1$.

$\sum_{n=0}^{\infty} ar^n$

The **geometric series test** states:

- If $|r| < 1$, the series converges.
- If $|r| \geq 1$, the series diverges.

Example:

$\sum_{n=0}^{\infty} \frac{1}{2^n}$

Here, $a = 1$ (first term) and $r = \frac{1}{2}$.

- Since $|r| = \frac{1}{2} < 1$, the series converges.
- The sum is given by:

$$\text{Sum} = \frac{a}{1-r} = \frac{1}{1-\frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2$$

$\sum_{n=1}^{\infty} \frac{1}{n^p}$

The **p-series test** states:

- If $p > 1$, the series converges.
- If $p \leq 1$, the series diverges.

Example:

$\sum_{n=1}^{\infty} \frac{1}{n^2}$

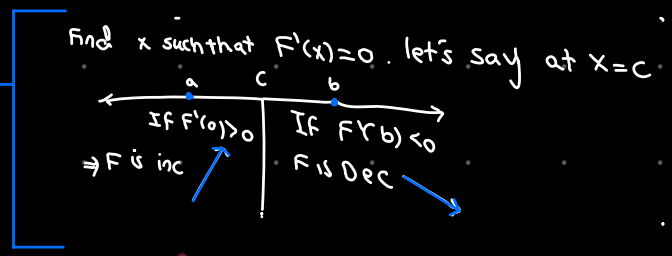
Here, $p = 2$.

- Since $p > 1$, the series converges.

11.3 (Integral Test)

Steps :

1. Check the Conditions: Ensure the function f(x) is:
   . Positive (f(x) > 0)
   . Decreasing (First derivative --> f '(x) < 0)
   . Continuous on [1, ∞)



2. Set Up the Improper Integral:

$$\sum a_n \longrightarrow \int_1^{\infty} f(x)\,dx$$

3. Evaluate the Integral:
   A- make ∞ --> t
   B- solve the integral
   C- take the limit

$$\lim_{t\to\infty} \int_1^{t} f(x)\,dx = ?$$

4. Determine Convergence or Divergence:

- The series $\sum a_n$ **converges** if the improper integral converges (finite result). → 0 & (1,2,3...)
- The series $\sum a_n$ **diverges** if the improper integral diverges (infinite result).

The series is:

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Let $f(x) = \frac{1}{x^2}$. Here's how the conditions are satisfied:

1. $f(x)$ **is positive:**

$f(x) = \frac{1}{x^2} > 0 \quad$ for all $x > 0$.

The function is always positive because the denominator $x^2 > 0$.

2. $f(x)$ **is continuous:**

$f(x) = \frac{1}{x^2} \quad$ is continuous for all $x > 0$.

There are no breaks, jumps, or undefined points for $x > 0$.

3. $f(x)$ **is decreasing:**

Compute the derivative of $f(x)$:

$f'(x) = \frac{d}{dx}\left(\frac{1}{x^2}\right) = -\frac{2}{x^3}$.

Since $f'(x) < 0$ for $x > 0$, the function is decreasing for $x > 0$.

Use it when terms resemble a function that is easier to integrate.

11.4 (Comparison & Limit Comparison Test)

Aspect	Direct Comparison Test	Limit Comparison Test
Key Idea	Directly compare a_n with b_n using inequalities.	Compare a_n and b_n using the ratio $\frac{a_n}{b_n}$.
Condition	Requires $a_n \leq b_n$ or $a_n \geq b_n$.	Requires the limit $L = \lim_{n\to\infty} \frac{a_n}{b_n}$.
Use Case	Works when it's easy to compare terms directly.	Works when direct comparison is difficult but a_n and b_n behave similarly.
Key Result	Behavior depends on $\sum b_n$ (converges/diverges).	Behavior depends on L and $\sum b_n$ (converges/diverges).

Steps: (depends on which test you take...)

1. Direct Comparison Test

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A. Find a known series $\Sigma\ b_n$ (Example --> $\Sigma\ 1/x^p$ or a geometric series).

B. Determine whether b_n is divergent or convergent

C. Compare a_n with b_n to establish:

- If $a_n \leq b_n$ and $\Sigma\ b_n$ converges, then $\Sigma\ a_n$ also converges.
- If $a_n \geq a_n$ and $\Sigma\ b_n$ diverges, then $\Sigma\ a_n$ also diverges.

2. Limit Comparison Test

The Limit Comparison Test refines the Direct Comparison Test by using the limit of the ratio between a_n and b_n.

A. Compute the limit of the ratio (a_n / b_n) :

$$L = \lim_{n\to\infty} \frac{a_n}{b_n}.$$

B. Use the value of L :

- $0 < L < \infty$:

  Both $\Sigma\ a_n$ and $\Sigma\ b_n$ either converge or diverge. (DEPENDS ON THE BEHAVIOUR OF $\Sigma\ b_n$)

- $L = 0$:

  If $\Sigma\ b_n$ converges, then $\Sigma\ a_n$ also converges.

- $L = \infty$:

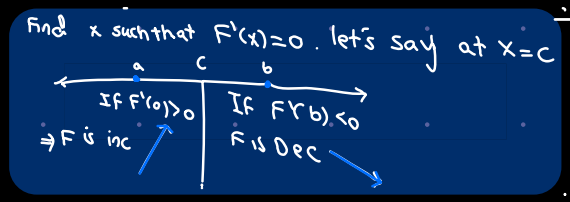
  If $\Sigma\ b_n$ diverges, then $\Sigma\ a_n$ also diverges.

11.5 (Alternating Test)

An alternating series is a series whose terms alternate in sign, typically of the form: $\sum_{n=1}^{\infty}(-1)^{n+1}b_n$ or $\sum_{n=1}^{\infty}(-1)^{n}b_n$

Conditions for Convergence

1- Decreasing Terms: $b_n \geq b_{n+1}$ (using the First derivative test or the graph to justify this condition)



2- $\lim_{n\to\infty} b_n = 0$

- If both conditions are met, the alternating series conditionally converges. and you can test the absolute series $\Sigma\,|a_n|$ to check if it absolutely converges

$\Sigma\,|a_n|$ Absolute Convergence Test --> checks if the series absolutely converges

- If $\Sigma\,|a_n|$ converges → The series is absolutely convergent.
- If $\Sigma\,|a_n|$ diverges → The series is conditionally convergent.

- If either condition is not met, the test fails, and the series may diverge. (you need another test)

Example to Illustrate

Case 1: Absolute Convergence

- Series: $\sum (-1)^n \frac{1}{n^2}$
- Absolute series: $\sum |a_n| = \sum \frac{1}{n^2}$, which converges (since $p = 2 > 1$).
- Conclusion: The series is **absolutely convergent** (no need to apply AST).

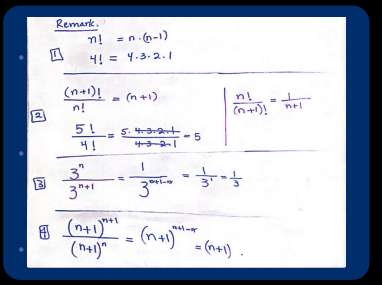
Case 2: Conditional Convergence

- Series: $\sum (-1)^n \frac{1}{n}$
- Absolute series: $\sum |a_n| = \sum \frac{1}{n}$, which diverges (harmonic series).
- Apply AST:
  - $b_n = \frac{1}{n}$ is decreasing.
  - $\lim_{n \to \infty} b_n = 0$.
- Conclusion: The series is **conditionally convergent**.

Case 3: Divergence

- Series: $\sum (-1)^n \frac{n}{n+1}$
- Absolute series: $\sum |a_n| = \sum \frac{n}{n+1}$, which diverges.
- Apply AST:
  - $b_n = \frac{n}{n+1}$ is not decreasing.
  - $\lim_{n \to \infty} b_n = 1 \neq 0$.
- Conclusion: The series **diverges**. ⟶ by using another test (divergence test)

11.6 (Ratio & Root Test)



{ Root Test }

Consider the series $\Sigma\, a_n$ and let:

$$L = \lim_{n\to\infty} \sqrt{|a_n|}$$

- If $L < 1$: The series converges absolutely.
- If $L > 1$ or $L = \infty$: The series diverges.
- If $L = 1$: The test is inconclusive. (use another test)

- Use the Root Test when terms involve powers (e.g., a_n = c^n or a_n = n^n).

Root Test Example:

Series:

$$\sum_{n=1}^{\infty} \frac{1}{n^n}$$

Test:

Let $a_n = \frac{1}{n^n}$. Compute:

$$L = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\frac{1}{n^n}}$$

$$L = \lim_{n\to\infty} \frac{1}{n} = 0$$

Result:

Since $L = 0 < 1$, the series converges **absolutely**.

{ Ratio Test }

Consider the series $\Sigma\, a_n$ and let:

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

- If $L < 1$: The series converges absolutely.
- If $L > 1$ or $L = \infty$: The series diverges.
- If $L = 1$: The test is inconclusive. (use another test)

- Use the Ratio Test when terms involve factorials (n! , n^n , 6^n , e^n) or other sequences with clear ratios.

Ratio Test Example:

Series:

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

Test:

Let $a_n = \frac{n!}{n^n}$. Compute the ratio:

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{\frac{(n+1)!}{(n+1)^{n+1}}}{\frac{n!}{n^n}} = \frac{(n+1)! \cdot n^n}{n! \cdot (n+1)^{n+1}}$$

Simplify:

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1) \cdot n^n}{(n+1)^{n+1}} = \frac{n^n}{(n+1)^n}$$

Taking the limit as $n \to \infty$:

$$\lim_{n\to\infty} \frac{n^n}{(n+1)^n} = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^n = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right)^n = \frac{1}{e}$$

Result:

Since $L = \frac{1}{e} < 1$, the series converges **absolutely**.

11.8 (Power Series)

A power series is a series of the form: $\sum_{n=0}^{\infty} c_n (x-c)^n$

Where:

- c_n are coefficients,
- x is the variable,
- c is the center of the series.

- The power series itself is not a test, but the Ratio Test or Root Test is often applied to find the radius and interval of convergence for a power series. We also need the center but it's already given in the question.

$$\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$$

Step 1: Apply the Ratio Test

Let $a_n = \frac{(x-2)^n}{n}$. Compute:

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{\frac{(x-2)^{n+1}}{n+1}}{\frac{(x-2)^n}{n}}\right|$$

Simplify:

$$L = \lim_{n\to\infty} \left|\frac{(x-2)^{n+1}}{n+1} \cdot \frac{n}{(x-2)^n}\right| = \lim_{n\to\infty} \left|\frac{(x-2)\cdot n}{n+1}\right|$$

$$L = \lim_{n\to\infty} |x-2| \cdot \frac{n}{n+1} = |x-2| \cdot 1 = |x-2|$$

Convergence Criterion:

The series converges if:

$|x-2| < 1$

So the **radius of convergence** is:

$R = 1$

Step 2: Find the Interval of Convergence

The interval based on the radius is:

$-1 < x-2 < 1 \quad \Rightarrow \quad 1 < x < 3$

Now, test the endpoints $x = 1$ and $x = 3$ **separately**:

1. **At $x = 1$:**

Substituting $x = 1$ into the series:

$$\sum_{n=1}^{\infty} \frac{(1-2)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

This is the **alternating harmonic series**, which converges.

2. **At $x = 3$:**

Substituting $x = 3$ into the series:

$$\sum_{n=1}^{\infty} \frac{(3-2)^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$$

This is the **harmonic series**, which diverges.

Make sure you are only substituting X not n

Final Answer:

The interval of convergence is:

$[1, 3)$

- The series converges at $x = 1$.
- The series diverges at $x = 3$.

Steps to Answer a Series Question

1. State the Series: Clearly write the series you are analyzing.

2. Choose and Apply the Test:
- Use the appropriate test (e.g., Ratio Test, Integral Test, P-Series Test).
- Show your work step-by-step.

3. State the Conclusion:
- Clearly say whether the series converges or diverges.

4. Mention the test you used.